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Measurement of Raman beam inclination changes in a cold atom gravimeter based on Raman beam interference

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The gravimeter inclination is a significant parameter for cold atom gravimeters, and the counter-propagating Raman beams should be exactly parallel to the local vector of gravity. The tiltmeters, essential devices in cold atom gravimeters, are used to determine the optimum inclination of Raman beams and compensate for the inclination error. However, the conventional tiltmeters may lead to system errors in cold atom gravimeters due to insufficient nonlinearity and drift. In this study, we establish an optical interferometer inside the cold atom gravimeter by placing a hollow beam splitter plate in the path of the Raman beams. This optical interferometer acts as a tiltmeter to measure the inclination change of the Raman beams without influencing the gravity measurement. We prove that our optical tiltmeter (OT) works well with field assembly. Comparisons of our OT and commercial tiltmeters reveal that the nonlinearity of our OT is at least one-tenth of that of the commercial tiltmeters, and that the drift of our OT is at least 23 μrad less than that of the commercial tiltmeters over 90 h measurements. This can reduce the typical value of the atom gravimeter system error by 4 μGal. Further, a comparison of measured gravity to inclination deviation calibrated our OT and further validated that our OT outperforms commercial tiltmeters. This work enables more precise measurement of Raman beam angle variations and facilitates the calibration of installed tiltmeters, whether in the laboratory or the field.

Keywords: Raman beam interference; Fourier transform; tilt measurement.

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1. Introduction

The cold atom gravimeter, which is a crucial tool for exploring and perceiving the Earth[1–5], offers high accuracy and a rapid sampling rate for continuous gravity measurements. The cold atom gravimeter can be deployed in various settings, including static environments[6,7], vehicle-mounted platforms[8], shipboard installations[9,10], airborne systems[11], and microgravity conditions[12–15]. Additionally, cold atom gravimeters play a significant role in fundamental physics, such as determining the Newtonian gravitational constant[16] and verifying the weak equivalence principle[17,18].

The Raman beam sequence splits, reflects, and recombines the atomic wave packet for interferometry[19], playing a critical role in gravity measurements. The Raman beams should be strictly parallel to the direction of gravity; otherwise, the deviation of the measured gravity is proportional to the square of the inclination deviation. To correct the systematic error caused by the inclination error, scholars have proposed numerous approaches[7,20–26]. In all cases, tiltmeters are required to monitor inclination change. In general, tiltmeters use MEMS devices or electrolytes to locate the direction of gravity and obtain the inclination variation. These devices are mounted on the frame of the Raman beam correlation optics to continuously monitor inclination change.

However, these conventional tiltmeters have some problems in use, such as the nonlinearity may not be good enough, which can lead to deviations from the measured optimum inclination, the presence of sensitive components within the sensor, stress relief, component aging, charge leakage, and changes in ambient temperature[27,28], leading to deviations from the compensated gravity value. This affects the precision and accuracy of the gravimeter[7,26]. For example, for a typical micro electro mechanical system (MEMS) tiltmeter (MT) used for a year of long-term measurements, the deviation may be up to 170 μrad, resulting in a change in the gravity value of more than 58 μGal. Electrolyte-based tiltmeters (ET) with better performance also

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have uncertainties in extinguishing the drift, and have questionable nonlinearity performance. In addition, the conventional tiltmeter measures the frame rather than the Raman beam. There are numerous angle measurement technologies such as optical, capacitive, Hall effect, inductive, and magneto-resistive technologies; most are used to measure the roll angle rather than the pitch angle and are sensitive to environmental influences[29,30]. This makes them unsuitable for cold atom gravimeters. A method borrowed from optical gravimetry was used to determine the verticality and measure the inclination change of Raman light using a position-sensing detector (PSD); the vertical calibration deviation of this method is less than 200 μrad, which corresponds to a systematic error of 20 μGal in gravity measurements, and an accuracy of 30 μrad for inclination change measurements[31]. This may not be sufficient for high-precision gravity measurements. Optical interferometry, which offers resolution and nonlinearity in the nano-radian scale[32], is one of the most accurate angular measurement methods. In this study, we present a novel Raman beam inclination change measurement method based on Raman beam interference for cold atom gravimeters.

The rest of this paper is organized as follows. In Sec. 2, we discuss how to construct an optical interference for an atom gravimeter without disturbing the gravity measurements and describe the principle of our method. In Sec. 3, we present the experimental procedure and results, including the field, nonlinearity, and drift tests, and the relationship between the gravimeter's inclination and the measured gravity. Finally, in Sec. 4, we summarize our work.

2. Theoretical Background and the Proposed Method

In the following, we provide a brief review of the gravity measurement bias caused by inclination measurement errors. The Raman beams of a typical cold atom gravimeter, which contain two frequency components (k_1, k_2), are collimated by the collimator, pass through the vacuum chamber and the quarter-wave plate in turn, and are then reflected by a retro-reflection mirror to form a wave vector in the opposite direction ($-k_1$, $-k_2$). The counter-propagating beams act on the atoms in the vacuum chamber to realize the interference of the atom wave packet. Together, k_1 and k_2 construct the effective wave vector of the Raman beam, denoted as k_{eff}. The relation between gravity and the Raman beam is given as $g = \Delta\Phi/k_{\text{eff}}T^2$, where $\Delta\Phi$ is the phase shift of the interferometer and T is the interrogation time between the Raman pulses.

If the two counter-propagating Raman beams always remain strictly overlapping, considering two Raman beams as a single unit, the measured gravity is a quadratic function of the inclination deviation of the gravimeter in an arbitrary one-direction space[25,26]:

$$g_\theta = g_0 - A(\theta_0 - \theta_C)^2, \tag{1}$$

where g_θ is the actual measured gravity, g_0 is the measured gravity without a tilt effect, i.e., measured gravity at optimum inclination, θ_0 is the optimum inclination, A is a constant factor, and θ_C is the actual inclination. According to Eq. (1), we can measure g_θ for different θ_C to determine θ_0 and A, and ensure that the Raman beam is strictly parallel to the direction of gravity. After setting the inclination to θ_0, the inclination of the Raman beams may drift due to the deformation, and thus, we need to compensate for the bias caused by the drift. Inclination measurement errors can lead to incorrect compensation. According to Eq. (1), if there is a measurement error θ_{error} in the measured value, i.e., the measured value is $\theta_C + \theta_{\text{error}}$, the gravity compensation error (g_{error}) caused by the inclination error (θ_{error}) can be expressed as

$$g_{\text{error}} = A[2(\theta_0 - \theta_C)\theta_{\text{error}} + \theta_{\text{error}}^2]. \tag{2}$$

In the case where the two Raman beams do not overlap, the gravity compensation error (g_{error}) caused by the inclination error of the retro-reflection mirror (θ_{error}) can be expressed as

$$g_{\text{error}} = 2A[2(\theta_0 - \theta_C)\theta_{\text{error}} + \theta_{\text{error}}^2]. \tag{3}$$

The term $(\theta_0 - \theta_C)$, which can be regarded as the drift or the inclination deviating from the optimum inclination, is usually in the tens of μrad range. The reason for multiplying by a factor of two is that the change in the angle of the reflected beam is twice the change in the angle of the mirror. The more tilt drift there is, the greater the gravity measurement error caused by the tilt measurement error.

2.1. Optical interference path setup inside an atom gravimeter

Introducing a beam splitter (BS) in the Raman beam path can easily construct an optical coherence system. However, to ensure that the gravity measurement process is not disturbed, careful consideration must be given to the placement of the BS. The addition of a BS alters the wavefront, polarization, and power of the Raman beam, and cold atom interferometers are highly sensitive to these three factors[33–36]. A polarizing beam splitter (PBS) can provide a polarized beam with sufficient quality. If the PBS is located between the vacuum chamber and the collimator, the wavefront aberrations present in both counter-propagating Raman beams will cancel and only aberrations present in one of the beams contribute to the phase error[33]. If the counter-propagating Raman beams do not strictly overlap, or the transmission wavefront of the PBS leads to diffusion or convergence, aberration may still exist. Placing a PBS between the vacuum chamber and the collimator appears to be a practical solution to fulfill our purpose and has been successfully applied in previous work[37]. However, PBS will obstruct the retro beam from entering the collimator since the polarizations of the counter-propagating Raman beams are mutually orthogonal. This prevents us from using the light reflected back into the fiber to optimize the inclination of the retro Raman beam[22]. Moreover, constructing such an optical interference path in a compact and

portable gravimeter, which is already in use, presents significant practical challenges.

The clear aperture of our Raman collimator is 24 mm, and approximately 10^8 atoms are captured; the temperature of the atoms is about 3 μK, and the interrogation time between the Raman pulses is 90 ms in our cold atom gravimeter. The diameter of the atom cloud may expand to about 1.3 cm when the final Raman pulse acts on the atoms. This indicates that only a portion of the Raman beam can interact with the atoms. We designed a hollow beam splitter plate based on this. As shown in Fig. 1, the hollow beam splitter utilizes the margin component of the Raman beam for optical interference and does not influence the middle component, which interacts with the atoms. Theoretically, the hollow BS can be placed anywhere along the Raman beam optical path. We place the BS between the quarter-wave plate and the retro-reflection mirror as it can quickly be disassembled and assembled in our system, and it is easier to match the optical path difference between the two optical interference arms.

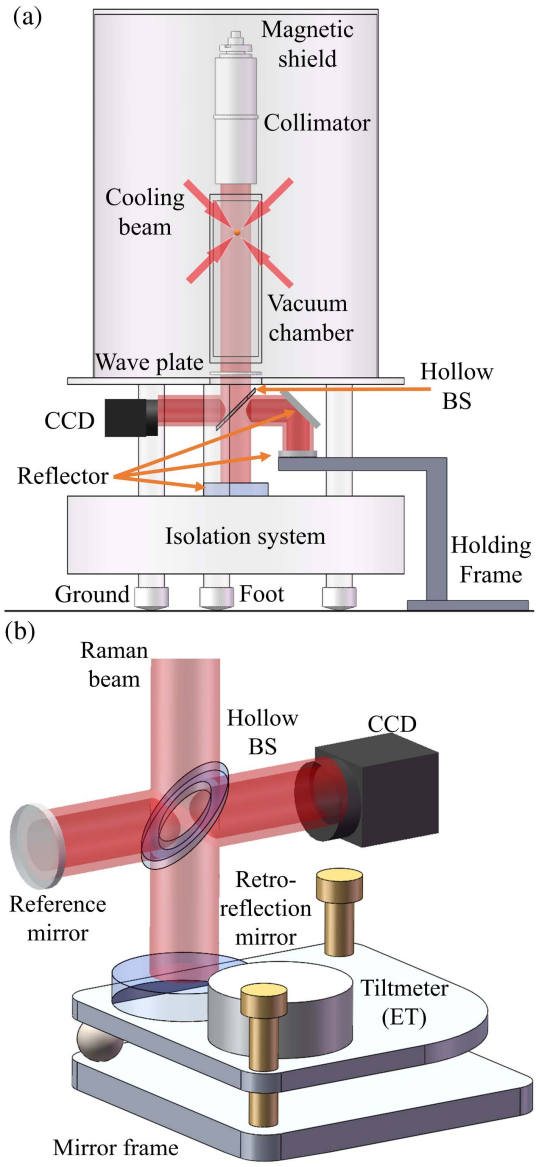


Fig. 1. Schematic diagram of the optical interference optical path inside the cold atom gravimeter. (a) Overall schematic and the setup for the whole gravimeter inclination measurement. (b) Specific schematic: the setup for retro-reflection Raman beam inclination measurements.

2.2. Measurement principle

The reference beam E_1 and the sample beam E_2 can be expressed as

$$\begin{cases} E_1(x,y) = a_1 e^{j2\pi(fx_1 \cdot x + fy_1 \cdot y)} \\ E_2(x,y) = a_2 e^{j2\pi(fx_2 \cdot x + fy_2 \cdot y)} \end{cases}, \tag{4}$$

and the interference equation can be written as

$$\begin{aligned} I(x,y) &= |E_2 + E_1|^2 \\ &= \{a_2^2 + a_1^2 + a_2 a_1 e^{j2\pi[(fx_2 - fx_1)\cdot x + (fy_2 - fy_1)\cdot y]} \\ &\quad + a_2 a_1 e^{-j2\pi[(fx_2 - fx_1)\cdot x + (fy_2 - fy_1)\cdot y]}\} \cdot S(x,y), \end{aligned} \tag{5}$$

where a_1 and a_2 are the amplifications of E_1 and E_2, respectively, and fx_1, fx_2, fy_1, and fy_2 are the space-frequencies of E_1 and E_2, respectively. As the interference fringes may have an irregular shape, $S(x,y)$ is taken as the function of the interferogram shape, with values of zero or one. If the tilt angle of the sample beam changes, the interference equation can be written as

$$\begin{aligned} I'(x,y) &= |E_2' + E_1|^2 \\ &= \{a_2^2 + a_1^2 + a_2 a_1 e^{j2\pi[(fx_2' - fx_1)\cdot x + (fy_2' - fy_1)\cdot y]} \\ &\quad + a_2 a_1 e^{-j2\pi[(fx_2' - fx_1)\cdot x + (fy_2' - fy_1)\cdot y]}\} \cdot S(x,y). \end{aligned} \tag{6}$$

Taking $I(x,y)$ as an example, as shown in Fig. 2, we now introduce the angle extracting process.

The pixels in the i row along the x-direction of the interferogram are taken out and subjected to a one-dimensional Fourier transform, and band-pass filtering is performed to get

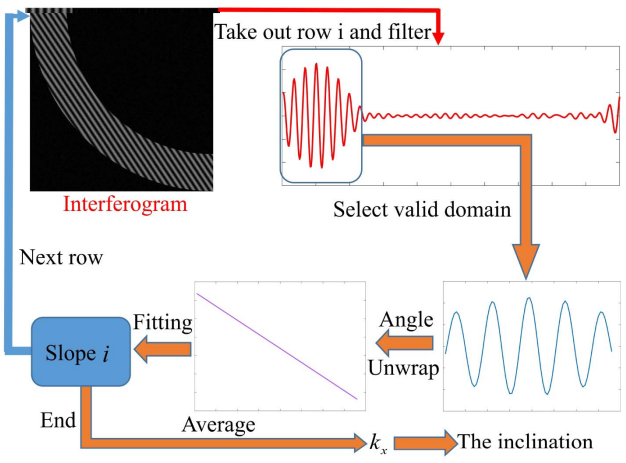


Fig. 2. Schematic diagram of the calculation process.

$$B(x, y_i) = a_2 a_1 e^{j2\pi[(fx_2 - fx_1)\cdot x]} \cdot S(x, y_i). \tag{7}$$

We retain the pixels that meet $|B(x, y_i)| > Q$, where Q is the threshold value. This process filters out the domains without interference fringes and extracts n_i pixels with interference fringes in row i. The phase angles are then extracted, unwrapping is performed to recover the continuous phase, and the resulting continuous phase is fitted using least-squares fitting to derive the spatial frequency ν_i of row i. The total spatial frequency along the x-direction is

$$fx_1 + fx_2 = \sum_i \nu_i n_i / \sum_i n_i = k_x, \tag{8}$$

which can be expressed as

$$\frac{\cos\ \alpha_1}{\lambda} + \frac{\cos\ \alpha_2}{\lambda} = k_x, \tag{9}$$

where α_1 is the angle between the reference beam and CCD, and α_2 is the angle between the sample beam and CCD. If the reference beam is perpendicularly incident to the CCD, the measurement angles can be written as

$$\alpha_2 = \arccos(k_x\lambda)/2, \qquad \beta_2 = \arccos(k_y\lambda)/2. \tag{10}$$

Using the same method, we can obtain the angle after changing in the x- and y-directions, denoted as α_2' and β_2', respectively. The variation of the inclination is

$$d\alpha = \alpha_2 - \alpha_2', \qquad d\beta = \beta_2 - \beta_2', \tag{11}$$

where $d\alpha$ and $d\beta$ are the angle variations along the x- and y-directions of the CCD, respectively.

If the reference beam is non-perpendicular to the CCD, the measurement angle in a single direction can be written as $\alpha_2 = \arccos[k_x\lambda + \cos(\alpha_1)]/2$, and the angle variation between $I(x, y)$ and $I'(x, y)$ is

$$\begin{aligned} d\alpha &= [\arccos(k_x\lambda + \cos\ \alpha_1) - \arccos(k_x'\lambda + \cos\ \alpha_1)]/2, \\ d\beta &= [\arccos(k_x\lambda + \cos\ \beta_1) - \arccos(k_x'\lambda + \cos\ \beta_1)]/2, \end{aligned} \tag{12}$$

where k_x, k_y, k_x', and k_y' are the measurement space-frequencies of the x- and y-directions of $I(x, y)$ and $I'(x, y)$.

Although we can obtain the angle variation over Eq. (11), we now discuss the impact of the non-perpendicular situation to make our approach more universal. We simulate the relationship among the measured angle variation, the actual angle variation, and α_1 based on Eq. (12).

As shown in Fig. 3, the less perpendicular the reference beam is to the CCD, the more the ratio varies with the measurement angle variation. However, the ratio variation is about 0.1% under a measurement range of 8 mrad even if the reference beam has an angular deviation of 0.45 rad. This indicates that the non-perpendicular situation has minimal impact on the cold atom gravimeter measurements. Therefore, the ratio can be treated as a constant and Eq. (12) can be rewritten as

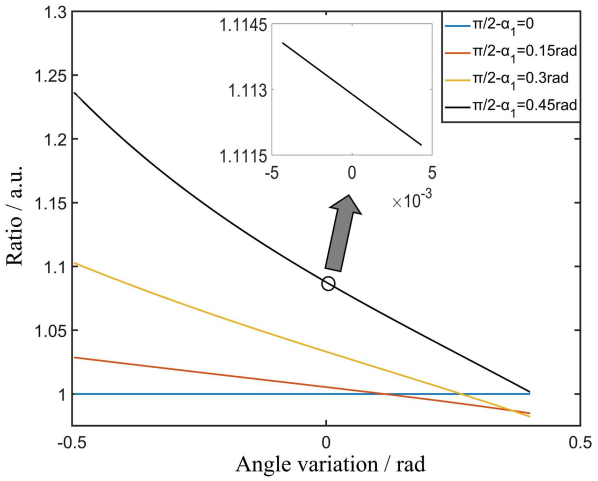


Fig. 3. Simulation of the relationship among the measured angle variation, the actual angle variation, and α_1. The ratio is the measured angle deviation divided by the true angle variation.

$$d\alpha = f_a \cdot d\alpha_A, \qquad d\beta = f_b \cdot d\beta_A, \tag{13}$$

where $d\alpha_A$ and $d\beta_A$ are the actual angle variations along the x- and y-directions, respectively, and f_a and f_b are the corresponding ratios.

3. Experiment

3.1. Field angle variation measurements compared with the commercial tiltmeter

We assembled our OT in the field and compared its performance to that of a commercial ET to verify the feasibility of the OT. The setup for this experiment is shown in Fig. 1(b). Specifically, we employed a Schäfter + Kirchhoff 60FC-Q780-4-M100S-37 collimator, a Jing Hang JHUM2000Bs-E CCD, and a Jewell 755 tiltmeter. The reference mirror, BS, and CCD were all fixed with the gravimeter. We adjusted the angle of the retro-reflection beam by adjusting the frame pitch of the Raman mirror. We obtained nine interferograms at different tilt angles and the corresponding outputs from the ET. A fluctuation of a few micro-radians in each direction was observed, which may be attributed to tiny vibrations. We confirmed that the gravity measurement remains unchanged before and after assembling our OT.

To illustrate that interferograms are highly contrasting and the shape of the interferogram is part of a ring that is difficult to process using conventional methods, one of the nine interferograms is depicted in Fig. 4. We process the interferograms using the method described in Sec. 2.2 and recombine all the rows together (bottom in Fig. 4). The fringes are much sharper after being processed by our method and retain the same shape as the raw interferogram. Comparisons between the top and bottom sides of Fig. 4 demonstrate that our method can effectively process the irregularly shaped interferogram.

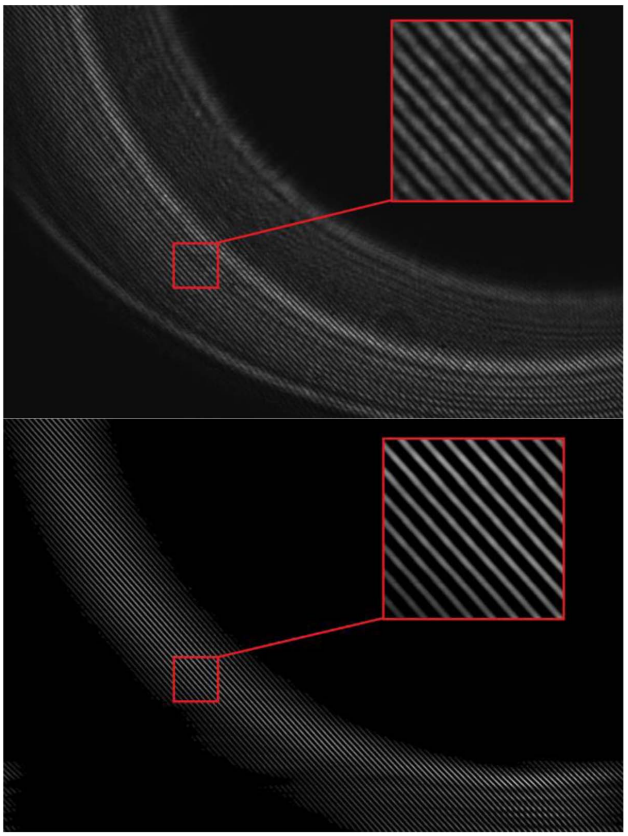


Fig. 4. A raw interferogram versus the image after filtering using our method. Top: original interferogram. Bottom: filtering with our method.

Calculating the difference between each measured inclinations and the starting inclination gives us nine values for the angle change in each direction. We denote $d\alpha_O$ as the angle variation measured by our OT in the x-direction and $d\beta_O$ as the angle variation in the y-direction. Moreover, the angle variation measured by the ET is denoted by $d\alpha_E$ and $d\beta_E$. The total angle variations are given as

$$d\theta_E = \sqrt{d\alpha_E^2 + d\beta_E^2}, \tag{14}$$

$$d\theta_O = \sqrt{d\alpha_O^2 + d\beta_O^2}, \tag{15}$$

where $d\theta_E$ and $d\theta_O$ are the total inclination variation measured by the ET and OT, respectively. For the non-perpendicular case, based on Eq. (15), we have

$$d\theta_c = \sqrt{(d\alpha_O/f_a)^2 + (d\beta_O/f_b)^2}, \tag{16}$$

where $d\theta_c$ is the corrected angle variation and f_a and f_b are the ratios described in Sec. 2.2. We assume that the measured inclination by the ET is precise and denote $d\theta_A$ as the actual angle variation; $d\theta_A$, $d\theta_E$, and $d\theta_O$ satisfy the following relationship:

$$d\theta_A = d\theta_O = d\theta_E, \tag{17}$$

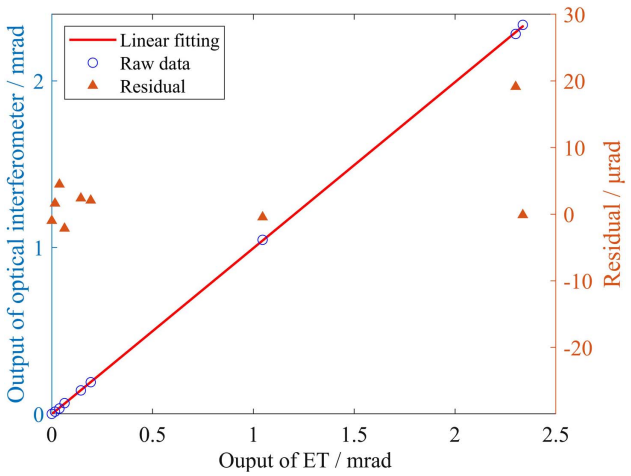


Fig. 5. Results of angular variation measurement in the field.

which can be expressed as

$$\sqrt{(d\alpha_O/f_a)^2 + (d\beta_O/f_b)^2} = \sqrt{d\alpha_E^2 + d\beta_E^2}. \tag{18}$$

Fitting the data with Eq. (18) yields f_a and f_b. To compare the measurement results from our OT with those of the ET, the residuals of the angular variation between the two measurements are calculated. The results are shown in Fig. 5. The trend of our OT measurements is consistent with that of the ET. This verifies that our OT is feasible even if we temporarily assemble it in the field, i.e., it is easy to implement and is robust. This experiment also provides a method for measuring the inclination of a retro-reflection Raman beam. However, an outlier residual datum of 19 μrad is significantly larger than the noise and other data. The error may originate from one of the tiltmeters or from operational mistakes.

3.2. Nonlinearity and drift

In this section, we compare the nonlinearity and drift of our OT with the ET and the MT to investigate the causes of the outlier datum mentioned in Sec. 3.1 and the performance of the OT. The ET and MT models are Jewell 755 and Ruifen ACA826T. The experimental setup is shown in Fig. 6. We fix the sample mirror, the conventional tiltmeters, and a small collimator on a rigid metal rod. The light from the small collimator hits the beam analyzer at a distance of about 1.8 m. We modulate the

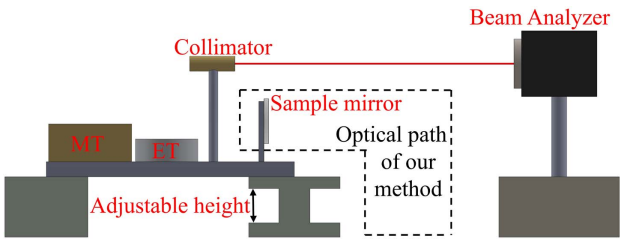


Fig. 6. Apparatus of the nonlinearity test.

inclination by adjusting the height of a fulcrum. The variation in angle is determined by monitoring the displacement of the spot on the beam analyzer. We consider that the accuracy of the beam analyzer is within a pixel, which is equivalent to about 1.9 μrad.

Considering the consistency of the tiltmeter's sensitive axis with the modulation direction, we calculated the coupling coefficient in the x- and y-directions for each tiltmeter and compensated for it. Linearly fitting the data provides us with the slope K of each tiltmeter. The residual is defined as $R = d\theta_{\text{tiltmeter}} - Kd\theta_{BA}$, where R is the residual, $d\theta_{\text{tiltmeter}}$ is the angle variation measured by the tiltmeters, $d\theta_{BA}$ is the reference angle variation measured by the beam analyzer, and θ_{BA} is the angular variation measured by the beam analyzer. This residual R is used to characterize the nonlinearity. We perform the experiment three times to eliminate the possibility of random factors, and the results are shown in Fig. 7. The residual curves of our OT appear to be random noise, while the residual curves of the conventional tiltmeters exhibit higher-order terms. The nonlinearity of our OT is approximately 0.4% within a measurement range of 2 mrad. In contrast, the data from the ET show a characteristic peak in all three experiments, with an amplitude of approximately 28 μrad and a range of 700 μrad. This characteristic peak can reasonably explain the outlier observed in Sec. 3.1. Therefore, within a measurement range of 2 mrad, the nonlinearity of the ET is approximately 1.4%, while the nonlinearity increases to around 4% if the measurement ranges within the interval of the characteristic peak.

We then placed the three tiltmeters statically for about 100 h to investigate their drift in a single direction. Due to the presence of tiny vibrations, we applied low-pass filtering with the same bandwidth to the data in order to better reveal the drift behavior. The results are presented in Fig. 8.

Figure 8 reveals that the measurement fluctuation of our OT remains below 7 μrad. In contrast, in the first 10 h, the ET measurements exhibit a drastic unidirectional change up to approximately 30 μrad, and subsequently stabilize. A notable drift of up to about 50 μrad is observed for the MT measurements. This cannot be attributed to long-term creep of the tiltmeter installation, as the long-term test was immediately followed by the nonlinear test of over a week's duration, during which the fixed tiltmeter was not reinstalled.

In summary, although the limitations of the measurement methodology prevent precise testing, our synchronous comparison of three tiltmeters demonstrates that our OT outperforms the conventional tiltmeter in both nonlinearity and drift. The relative nonlinearity of our OT is approximately 0.4% when the angle change range is under 2 mrad, while the relative nonlinearity of the ET is around 4.3% under a specific measurement range of 0.7 mrad. During over 90 h of measurements, the drift of our OT method remains below 7 μrad, which is at least 23 μrad lower than that of the conventional tiltmeters. According to Eq. (3), if the actual inclination drifts by 50 μrad in each direction or exhibits equivalent deviations from the optimum inclination, the combined effects of nonlinearity error and

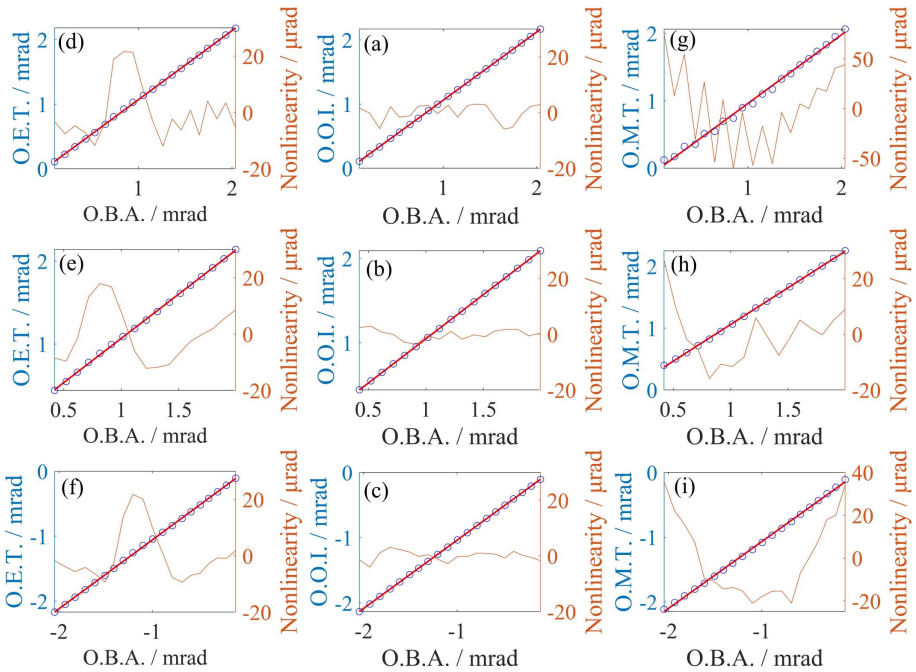


Fig. 7. Nonlinearity test results. O.O.I. is the output of our optical interference method; O.E.T. is the output of the ET; O.M.T. is the output of the MT; and O.B.A. is the output of the beam analyzer. The blue circles represent the measured data. The red solid lines are the linearly fitted results of the output of the tiltmeters and the beam analyzer. The orange curves represent the residual. The first set of results [(a), (d), and (g)] has been averaged over 20 s. The second and third sets of results [(b), (e), and (h); (c), (f), and (i)] have been averaged over 50 s to reduce the effect of small vibrations on the results.

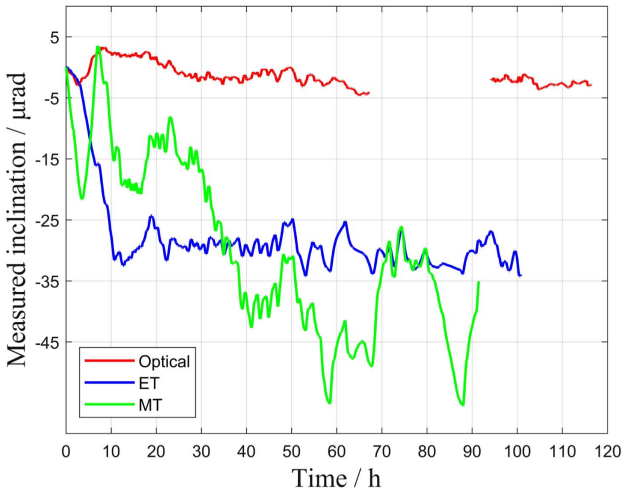


Fig. 8. Long-term test results. Storage issues resulted in the loss of some data.

the 23 μrad drift in the inclination measurement of the retro-reflection mirror could result in a total systematic error of approximately 4 μGal in the atom gravimeter.

3.3. Variation of measured gravity with inclination deviation

In this section, we investigate the relationship between gravity measurements and inclination measurements from our OT and the ET. We modulate the inclination of the whole gravimeter in this experiment. We keep one direction of the whole gravimeter unchanged while altering another orthogonal direction to obtain the x- and y-directions of the tilt measurements as they vary with the gravity measurements. The data are fitted using Eq. (1). The inclination of the unchanged direction is close to the optimum inclination (within 20 μrad), and thus the peak value of the fitted result of the two directions should be identical and close to g_0.

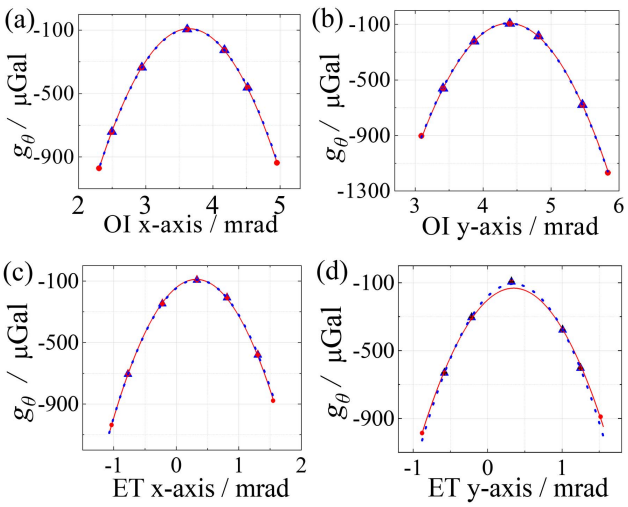


Fig. 9. Variation of measured gravity with inclination measurement. The blue triangles and dashed curves represent the five data points and the corresponding fitting results, respectively. The red circles are the appended points and the red solid curves are the fitting results of the seven points.

Table 1. Fitted Data of the Results in Fig. 9.

	g_0/μGal	A/a.u.	θ_0/mrad
OI-x blue dash	−92.5 ± 2.5	479.3 ± 3.5	3.6468 ± 0.0025
OI-y blue dash	−91.9 ± 3.1	493.2 ± 4.4	4.3566 ± 0.0030
OI-x red solid	−90.5 ± 2.4	484.2 ± 2.1	3.6434 ± 0.0015
OI-y red solid	−91.1 ± 4.8	493.3 ± 4.8	4.3513 ± 0.0038
ET-x blue dash	−91.8 ± 2.6	507.7 ± 3.9	0.3270 ± 0.0023
ET-y blue dash	−92.5 ± 2.5	613.2 ± 13.2	0.3450 ± 0.0042
ET-x red solid	−83.2 ± 5.2	515.7 ± 5.0	0.3199 ± 0.0033
ET-y red solid	−131.7 ± 12.3	562.2 ± 11.5	0.3509 ± 0.0054

To measure the tilt of the entire gravimeter using our OT, the reference mirror should be independent of the gravimeter. The configuration of this experiment is shown in Fig. 1(a). The BS and the CCD are fixed with the gravimeter. Unlike the setup described in Sec. 3.1, we add an additional 45° mirror, also fixed with the gravimeter. The reference mirror is fixed on a frame that is anchored to the ground. It is worth noting that the reference mirror cannot be parallel to the vector of gravity, as this would result in the loss of one measurement direction. We change the inclination of the whole gravimeter by adjusting the height of the feet.

The results are presented in Fig. 9 and Table 1. Although the five-point data ensure the fit, we append two additional points with larger inclinations of the gravimeter to further verify the nonlinearity. Before and after appending these two points, the two fitting curves from our OT nearly overlap, and the errors in the fitting parameters are within tolerance, while in the y-direction. In contrast, the two fitted curves in the y-direction of ET are clearly separated, the fitted data are highly variable, and g_0 is far from the actual value. This further confirms that our OT outperforms the ET and demonstrates that our OT can measure the inclination variations and determine the optimum inclination for the whole cold atom gravimeter. This experiment is also the calibration of the OT against the gravimeter, and after calibration, our optical interferometer can be used as an absolute tiltmeter.

4. Conclusion

We utilize the inherent Raman beam to construct an optical interferometer inside the gravimeter to measure the inclination change of the Raman beam. Ensuring no effect on the cold atom interference, a hollow BS is used to realize optical division

multiplexing. We propose a method based on one-dimensional Fourier filtering to process the irregular interferogram caused by hollow BS and extract valid information. The nonlinearity of our OT is 0.4%, while the nonlinearity of the commercial tiltmeters is over 4%, and our OT also has at least 23 μrad less drift than commercial tiltmeters in long-term measurements. This could result in a 4 μGal reduction in the systematic error of cold atom gravimeters when the actual tilt angle deviates from the optimal tilt angle by 50 μrad. In conclusion, our proposed optical tiltmeter for gravimeters is robust and has high accuracy, a simple setup process, and small drift, which makes it a reliable method to measure the inclination angle of gravimeters in the laboratory and field. More importantly, Raman beam interference may be a potential tool to study cold atom interferometers and improve their performance.

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References

1. M. Van Camp, O. de Viron, A. Watlet, *et al.*, "Geophysics from terrestrial time-variable gravity measurements," Rev. Geophys. **55**, 938 (2017).
2. B. Fores, C. Champollion, N. Le Moigne, *et al.*, "Assessing the precision of the iGrav superconducting gravimeter for hydrological models and karstic hydrological process identification," Geophys. J. Int. **208**, ggw396 (2016).
3. J. Kennedy, T. P. A. Ferré, and B. Creutzfeldt, "Time-lapse gravity data for monitoring and modeling artificial recharge through a thick unsaturated zone," Water Resour. Res. **52**, 7244 (2016).
4. M. Van Camp, O. de Viron, H.-G. Scherneck, *et al.*, "Repeated absolute gravity measurements for monitoring slow intraplate vertical deformation in western Europe," J. Geophys. Res. Solid Earth **116**, B8 (2011).
5. A. J. Romaides, J. C. Battis, R. W. Sands, *et al.*, "A comparison of gravimetric techniques for measuring subsurface void signals," J. Phys. D Appl. Phys. **34**, 433 (2001).
6. M. Kasevich and S. Chu, "Measurement of the gravitational acceleration of an atom with a light - pulse atom interferometer," Appl. Phys. B **54**, 321 (1992).
7. A. Peters, K. Y. Chung, and S. Chu, "High-precision gravity measurements using atom interferometry," Metrologia **38**, 25 (2001).
8. Q.-X. Li, X. Zhang, H.-K. Zhang, *et al.*, "Multi-scene mobile absolute gravity surveying by developing a vehicle-mounted static-base atom interferometry gravimeter," Measurement **231**, 114556 (2024).
9. Y. Bidel, N. Zahzam, C. Blanchard, *et al.*, "Absolute marine gravimetry with matter-wave interferometry," Nat. Commun. **9**, 627 (2018).
10. B. Wu, C. Zhang, K. Wang, *et al.*, "Marine absolute gravity field surveys based on cold atomic gravimeter," IEEE Sens. J. **23**, 24292 (2023).
11. Y. Bidel, N. Zahzam, A. Bresson, *et al.*, "Absolute airborne gravimetry with a cold atom sensor," J. Geod. **94**, 20 (2020).
12. F. Sorrentino, K. Bongs, P. Bouyer, *et al.*, "The space atom interferometer project: status and prospects," J. Phys. Conf. Ser. **327**, 012050 (2011).
13. G. M. Tino, F. Sorrentino, D. Aguilera, *et al.*, "Precision gravity tests with atom interferometry in space," Nucl. Phys. B Proc. Suppl. **203**, 243 (2013).
14. F. Migliaccio, M. Reguzzoni, K. Batsukh, *et al.*, "MOCASS: a satellite mission concept using cold atom interferometry for measuring the Earth gravity field," Surv. Geophys. **40**, 1029 (2019).
15. C. Pelluet, R. Arguel, M. Rabault, *et al.*, "Atom interferometry in an Einstein Elevator," arXiv preprint, arXiv:2407.07183 (2024).
16. G. Rosi, F. Sorrentino, L. Cacciapuoti, *et al.*, "Precision measurement of the Newtonian gravitational constant using cold atoms," Nature **510**, 518 (2014).
17. N. Gaaloul, H. Ahlers, T. A. Schulze, *et al.*, "Quantum tests of the equivalence principle with atom interferometry," Acta Astronaut. **67**, 1059 (2010).
18. D. Schlippert, J. Hartwig, H. Albers, *et al.*, "Quantum test of the universality of free fall," Phys. Rev. Lett. **112**, 203002 (2014).
19. M. Kasevich and S. Chu, "Atomic interferometry using stimulated Raman transitions," Phys. Rev. Lett. **67**, 181 (1991).
20. P. A. Altin, M. T. Johnsson, V. Negnevitsky, *et al.*, "Precision atomic gravimeter based on Bragg diffraction," New J. Phys. **15**, 023009 (2013).
21. A. Senger, "A mobile atom interferometer for high-precision measurements of local gravity," Humboldt-Universität zu Berlin, Mathematisch - Naturwissenschaftliche Fakultät I (2012).
22. M. Hauth, C. Freier, V. Schkolnik, *et al.*, "First gravity measurements using the mobile atom interferometer GAIN," Appl. Phys. B **113**, 49 (2013).
23. Q. Bodart, S. Merlet, N. Malossi, *et al.*, "A cold atom pyramidal gravimeter with a single laser beam," Appl. Phys. Lett. **96**, 13 (2010).
24. V. Ménoret, P. Vermeulen, N. Le Moigne, *et al.*, "Gravity measurements below 10−9 g with a transportable absolute quantum gravimeter," Sci. Rep. **8**, 12300 (2018).
25. B. Wu, B. Cheng, Z.-J. Fu, *et al.*, "Measurement of absolute gravity based on cold atom gravimeter at large tilt angle," Acta Phys. Sin. **65**, 060302 (2016).
26. H.-T. Xie, B. Chen, J.-B. Long, *et al.*, "Tilt adjustment for a portable absolute atomic gravimeter," Chinese Phys. B **29**, 073701 (2020).
27. Y. K. Cheung and H. Yu, "Electrochemical tilt sensors with symmetric concentric electrode pairs," IEEE Sens. J. **21**, 21322 (2021).
28. R. Matsuda, S. Zihao, U. Kamoto, *et al.*, "Liquid-based digital readable tilt sensor," Adv. Mater. Technol. **6**, 2100490 (2021).
29. A. S. Anil Kumar, B. George, and S. C. Mukhopadhyay, "Technologies and applications of angle sensors: a review," IEEE Sens. J. **21**, 7195 (2020).
30. S. Wang, R. Ma, F. Cao, *et al.*, "A review: high-precision angle measurement technologies," Sensors **24**, 1755 (2024).
31. H.-Q. Luo, Y.-Y. Xu, J.-F. Cui, *et al.*, "Determining the tilt of the Raman laser beam using an optical method for atom gravimeters," Chinese Phys. B **33**, 123701 (2024).
32. Z. Ge and M. Takeda, "High-resolution two-dimensional angle measurement technique based on fringe analysis," Appl. Opt. **42**, 6859 (2003).
33. V. Schkolnik, B. Leykauf, M. Hauth, *et al.*, "The effect of wavefront aberrations in atom interferometry," Appl. Phys. B **120**, 311 (2015).
34. A. Louchet-Chauvet, T. Farah, Q. Bodart, *et al.*, "The influence of transverse motion within an atomic gravimeter," New J. Phys. **13**, 065025 (2011).
35. D. S. Weiss, B. C. Young, and S. Chu, "Precision measurement of $\hbar$/mCs based on photon recoil using laser-cooled atoms and atomic interferometry," Appl. Phys. B **59**, 217 (1994).
36. J. Le Gouët, "Étude des performances d'un gravimètre atomique absolu: sensibilité limite et exactitude préliminaire," Université Paris Sud - Paris XI (2008).
37. Y. Zhao, S. Wang, W. Zhuang, *et al.*, "Raman-laser system for absolute gravimeter based on 87Rb Atom interferometer," Photonics **7**, 32 (2020).